MACHINE LEARNING PROJECT

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Course / Section: BSIS 4A

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MILESTONE 1 (WORKFLOW STAGE A)

A. Problem Definition: Market Basket Analysis Using Association Rule Mining

Brief Description and Problem Statement

Brief Description

Businesses collect large volumes of historical retail transaction data, yet raw receipt logs alone do not show which items customers frequently buy together. This project applies Market Basket Analysis using Association Rule Mining (specifically the Apriori algorithm) to uncover meaningful product co-occurrence patterns from the Online Retail Excel dataset (Online Retail.xlsx). The discovered association rules will support data-driven business decisions in product placement, promotions, product bundling, cross-selling, and inventory planning.

Problem Statement

Using 541,909 historical e-commerce transaction line items from the Online Retail Excel dataset (December 2010 to December 2011), this project applies the Apriori Association Rule Mining algorithm to identify frequent product combinations and association rules that outperform a random guessing baseline (Lift greater than 1.0), enabling businesses to optimize product placement, promotions, bundling, and cross-selling.

Dataset and Scope Requirements

Ownership, Source, and License: This project uses one single dataset: the Online Retail Data Set (Online Retail.xlsx), originally donated by Dr. Daqing Chen of London South Bank University to the UCI Machine Learning Repository (Dataset ID: 352) and accessed via Kaggle at https://www.kaggle.com/datasets/jihyeseo/online-retail-data-set-from-uci-ml-repo. The dataset is provided in Microsoft Excel (.xlsx) format, allowing direct viewing in Excel as well as through Kaggle's browser preview, and is licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license for academic and analytical use with proper citation.

Dataset Size and Usable Features: The Excel file contains 541,909 transaction line-item rows across 8 columns (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country). When grouped by InvoiceNo, these records represent approximately 25,900 unique shopping transactions across roughly 4,070 distinct products. This comfortably exceeds the minimum project requirement of at least 500 rows and 5 usable features.

Privacy and Anonymization: The dataset contains no personal or sensitive information such as customer names, home addresses, email addresses, or payment details. Customers are identified only by a numeric CustomerID (which is missing in 24.9 percent of guest checkout rows). Because Market Basket Analysis groups items strictly by InvoiceNo, the CustomerID column will be dropped completely prior to modeling so that no individual customer can be tracked or re-identified. Only aggregated association rules and a small anonymized sample of transactions will be shared in the final deliverables.

Project Scope: To ensure the project is thorough, reproducible, and completed cleanly within the semester timeline, the scope focuses on building a clean Association Rule Mining pipeline using the Apriori algorithm on this single Excel dataset. Transactions will be cleaned (removing cancelled invoices and invalid quantities or prices), split into training and testing sets, mined for frequent itemsets and association rules, and evaluated using Support, Confidence, and Lift.

A1. What real-world problem does your dataset address, and who benefits from a solution?

Real-World Problem: Retail and e-commerce organizations accumulate large amounts of transaction data every day, but raw invoice tables do not reveal which products customers routinely purchase together within the same shopping basket. Without an automated way to uncover these purchasing relationships, businesses struggle to plan store layouts, group complementary products, or design effective promotions.

Who Benefits from a Solution:

1. Retail and E-Commerce Merchandising Teams: Can place frequently associated products near each other in physical stores or online catalogs and create data-backed product bundles.

2. Marketing and Sales Teams: Can design targeted promotions and cross-selling recommendations (suggesting related items when a customer buys a primary item) instead of relying on guesswork.

3. Inventory and Supply Chain Planners: Can better understand correlated product demand so that linked items are stocked together before peak buying periods.

4. Customers: Benefit from more relevant product recommendations and convenient product bundles tailored to their shopping habits.

A2. Is this classification, regression, clustering or something else? Why does that framing fit the problem?

Chosen Framing: This project uses Association Rule Mining (also known as Pattern Mining), which falls under the 'something else' category of unsupervised machine learning.

Why This Framing Fits the Problem: Our dataset is a collection of customer shopping receipts showing which products were bought in each transaction. There is no single answer column to predict (so it is not classification or regression), and we are not trying to sort customers into fixed groups (so it is not clustering). Instead, our goal is to find 'if-then' shopping patterns—for example, discovering that if a customer puts Bread and Butter in their basket, they are also likely to buy Jam. Association Rule Mining is built specifically to scan thousands of receipts and find these exact product-to-product buying patterns.

A3. What is your target (or outcome of interest) and how was it measured or labelled?

Outcome of Interest: Because Association Rule Mining is unsupervised, there is no single target column or ground-truth label in the raw dataset. Instead, the outcome of interest is a set of frequent itemsets and directional association rules (each linking an 'if' product set to a 'then' product set) representing products frequently bought together within the same transaction.

How It Is Measured and Structured: Raw transaction line items are cleaned to remove cancelled invoices and non-positive quantities or unit prices, then grouped by the transaction identifier (InvoiceNo). Each transaction is converted into a simple yes-or-no basket table (1 if the

product was purchased in that invoice, and 0 if it was not). Association rules are then generated from this table and filtered using minimum thresholds for Support, Confidence, and Lift.

A4. Which metric(s) define success, and why do they match the real cost of different errors?

Specific Metrics Used: To measure whether a discovered product rule is actually useful, we use three specific Association Rule Mining metrics: Support, Confidence, and Lift.

1. Support (How popular is the combo?): Support measures how often two items are bought together out of all receipts in the store. For example, if 100 people shop at the store and 10 of them buy both Bread and Jam together, the Support is 10 out of 100 (10 percent).

2. Confidence (How reliable is the rule?): Confidence measures how often customers buy the second item when they have already picked up the first item. For example, if 20 people buy Bread, and 10 of those Bread buyers also grab Jam, the Confidence is 10 out of 20 (50 percent).

3. Lift (Is it a real connection or just luck?): Lift tells us if buying the first item actually makes someone more likely to buy the second item, compared to normal everyday shopping. A Lift of 1.0 means there is no real connection (just random coincidence). A Lift greater than 1.0 means the items truly belong together (for example, a Lift of 2.5 means customers who buy the first item are 2.5 times more likely to buy the second item than usual).

Why These Metrics Match Real Business Costs: In a real store, making bad product recommendations costs money in two ways. First, if a store creates a special promotion for two items that almost nobody ever buys together, the store wastes marketing money and shelf space—setting a minimum Support score prevents this mistake by ensuring enough customers actually want the combo. Second, if a store offers a discount bundle on an item that everyone already buys anyway (like a top-selling gift bag), the store loses profit by giving away unnecessary discounts—requiring a high Lift score (greater than 1.0) prevents this mistake by proving that the first item genuinely encourages customers to buy the second item.

A5. Where did the data come from (source, collection method, size, time period, licence)?

Source: The dataset is the Online Retail Data Set (Online Retail.xlsx) from the UCI Machine Learning Repository (Dataset ID: 352), donated by Dr. Daqing Chen of London South Bank University, and mirrored on Kaggle.

Collection Method: The records were collected directly from the transactional database of a UK-based, non-store online retail company that primarily sells unique all-occasion giftware, with many customers operating as wholesalers.

Dataset Size: The Excel file contains 541,909 transaction line-item rows and 8 attributes (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country). When grouped by InvoiceNo, these records form approximately 25,900 unique transactions across roughly 4,070 distinct products.

Time Period: All recorded transactions took place between December 1, 2010 and December 9, 2011 (one full year of retail operations).

License: The dataset is released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license via the UCI Machine Learning Repository, permitting academic and analytical use with proper citation.

A6. Are there privacy, consent or fairness concerns with this data or its use?

Privacy and Anonymization: The dataset does not contain personally identifiable information such as customer names, street addresses, email addresses, phone numbers, or payment card numbers. Customers are represented only by an anonymized 5-digit CustomerID, which is missing in roughly 24.9 percent of rows due to guest checkouts. Since Market Basket Analysis groups items strictly by InvoiceNo, CustomerID will be removed completely before analysis to eliminate any re-identification risk.

Consent and Ownership: The dataset was publicly released through the UCI Machine Learning Repository under an open CC BY 4.0 license specifically for research and educational use.

Fairness and Representation Concerns: Because the records come from a single UK online giftware company during 2010–2011 and include bulk wholesale orders alongside individual retail shoppers, the discovered rules reflect UK holiday shopping trends and wholesale buying behavior. Therefore, these rules should not be applied blindly to unrelated retail industries or regions without retraining on local transaction data.

A7. What score would a naive baseline achieve (for example majority class or predicting the mean)?

What a Naive Baseline Means Here: A naive baseline is the simplest possible 'no-skill' guess we could make before using machine learning. In Market Basket Analysis, the naive baseline is 'Random Guessing'—assuming that what a customer puts in their basket has zero effect on what else they buy.

Simple Everyday Example: Imagine 100 shoppers walk into a store, and 20 of them (20 percent) normally buy Milk regardless of what else they are shopping for. Now imagine 10 shoppers put a Toothbrush in their basket. If Toothbrushes and Milk have no real connection, how many of those 10 Toothbrush buyers will also buy Milk just by pure chance? The same 20 percent (2 out of 10 people).

The Exact Score of This Naive Baseline:

1. Naive Baseline Lift = 1.00: Because the 20 percent buying rate with a Toothbrush is the exact same as the normal 20 percent buying rate without a Toothbrush (20 percent divided by 20 percent = 1.00), random guessing always gets a Lift score of 1.00.

2. Naive Baseline Confidence: If a store uses a simple 'always suggest the most popular item' rule—recommending the single best-selling product in the store to every shopper—its Confidence score is simply how popular that single item is overall (around 8 to 11 percent in our dataset), and its Lift score is still 1.00.

How We Beat the Baseline: For our Apriori machine learning model to be useful, the rules it discovers must beat random guessing on unseen test receipts by achieving a Lift score strictly greater than 1.00 (for example, a Lift of 2.0 or 3.0).

Dataset Link, Sample Preview, and References

https://www.kaggle.com/datasets/jihyeseo/online-retail-data-set-from-uci-ml-repo

(Official UCI Repository DOI: https://doi.org/10.24432/C5BW33).

Sample Rows from Online Retail.xlsx (First 5 Records):

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850	United Kingdom

References:

[1] D. Chen, "Online Retail," UCI Machine Learning Repository, 2015. [Online]. Available: https://doi.org/10.24432/C5BW33

[2] J. Seo, "Online Retail Data Set from UCI ML Repo (Online Retail.xlsx)," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/jihyeseo/online-retail-data-set-from-uci-ml-repo

[3] D. Chen, S. L. Sain, and K. Guo, "Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining," Journal of Database Marketing & Customer Strategy Management, vol. 19, no. 3, pp. 197–208, Sep. 2012.

[4] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules in large databases," in Proc. 20th Int. Conf. Very Large Data Bases (VLDB), Santiago, Chile, Sep. 1994, pp. 487–499.